

Seek: A Telehealth System for Preventing Age-Related Vision Loss Inspired by the Quantified Self

Melissa M.F. Li

School of Information Studies, McGill University, Montreal, Quebec
melissa.li2@mail.mcgill.ca

ABSTRACT

Applications of telehealth technologies have largely focused on management of chronic illnesses. However, the use of telehealth for preventative healthcare is becoming increasingly relevant as the aging population rapidly increases. In this paper we draw correlations between Quantified Self enthusiasts and those looking to monitor personal data for preventative healthcare. From this relation, we leverage aspects of the Quantified Self in order to define design requirements for preventative telehealth tools and present a telehealth system designed to aid prevention of age-related vision loss.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**;

KEYWORDS

Quantified Self, Telehealth, Older Adults, Preventative Healthcare, Vision loss

ACM Reference Format:

Melissa M.F. Li. 2021. Seek: A Telehealth System for Preventing Age-Related Vision Loss Inspired by the Quantified Self. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451519>

1 INTRODUCTION

In 2019, the World Health Organization (WHO) reported that 171 million adults over the age of 65 experience eyesight impairment ranging from severe vision loss to complete blindness [31]. Ninety percent of all vision loss cases are preventable [1]; however, factors such as availability, accessibility and affordability can prevent patients from receiving proper care. Fortunately, rising public interest in health and wellness technologies has spurred efforts to shift focus from curative to preventative healthcare. This paradigm shift meaningfully supports the development of telehealth technologies to address healthcare accessibility concerns, as well as support for the mitigation of age-related vision loss. Considering projections that the demographic of older adults will double by 2050 [29], the

design and development of preventative health technologies has become increasingly critical.

Literature from the HCI community that studies telehealth and self-managed healthcare for older adults can support innovation of preventative health technologies. The efficacy of telehealth for preventative healthcare has been established [22]; however, most of the telehealth research surrounding design and development concentrates on the management of chronic conditions, despite the differing motivations for adopting chronic telehealth technologies versus preventative telehealth technologies. Patients with chronic conditions often adopt telehealth tools and maintain self-tracking as part of a treatment plan, often predetermined with their doctor. Conversely, while preventative telehealth is also mediated by a health professional, it is more patient-driven and often results in a relatively diminished sense of urgency for the user. In this sense, we see that the demographic interested in preventative health technologies are more similar to the self-trackers of the Quantified Self movement, than to telehealth adopters with chronic health conditions. While people who use telehealth technologies are health focused in general, those using it for preventative measures require an intrinsic motivation, ambiguously guided intent and unsolicited participation that is seen in Quantified Selfers. These characteristics are important because they eminently influence system design and development. In this paper, we aim to leverage the research on the Quantified Self to facilitate understandings of how preventative telehealth systems should be designed. We apply this understanding and present a telehealth system dedicated to preventing age-related vision loss. Our research aims to stimulate conversation on redefining the course of HCI research on preventative self-management for health conditions and telehealth through the lens of the Quantified Self movement.

We designed Seek, a telehealth system that aims to support older adults in self-managing their vision health, thereby assuaging barriers often faced when seeking traditional vision care. While the main function of this system is to pair the mobile application with our ocular device (Figure 1) for ocular image capture and upload, this paper chiefly focuses on the design requirements and features of the mobile application. In the following sections of the paper, we highlight related literature, define design requirements, describe our design of Seek, and discuss future use cases of the system and broader opportunities of use.

2 RELATED WORK

2.1 The Quantified Self

The Quantified Self is a movement and cultural phenomenon where people voluntarily and autonomously collect and share information about their body vitals and daily habits through self-tracking technologies [26]. Rapid development of wearables such as Fitbit, Apple

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451519>



Figure 1: A 3D render of our supporting ocular device.

Watch and other biosensing technologies have supported this movement. Through ubiquitous computing capabilities, most systems are designed to work seamlessly with corresponding smartphone applications that extend to provide motivational support, a sense of community and aggregation of personal data, often presented with visualization features [1] [10]. A movement that spurred from mere leisurely interest in improving self-knowledge through personal quantification, has elicited a powerful audience of life hackers and academic researchers alike. Prior work has aimed to identify where the inherent value of self-tracking personal information lies [13], and mainly attributed it to facilitating achievement of user goals of behavioral change [3] [19] [20] [26]. Studies by Epstein et al. assert the notion that self-tracking is not solely adopted because of the benefits it derives for behavior change, but also due to curiosity, rewards and the enmeshment in users' everyday lives [12]. This idea of an unobtrusive data collection can explain why Quantified Self discourse is overwhelmingly committed to applications of passive homeostatic data collection such as heart rate, sleep and stress [8] [30].

While self-tracking technologies have proven to be useful, they are susceptible to pitfalls associated with routine or prolonged use. Reasons for user abandonment include: (1) tracking fatigue leading to incomplete datasets from collecting more data than necessary, (2) fragmented data scattered across multiple platforms, and (3) issues with preserving initial motivation [9] [10] [16] [17] [30]. Work by Choe et. al suggests that including initiating check-ins, simplifying the data upload/input process, making the self-tracking experience rewarding and keeping data management/exploration straightforward could mitigate and/or solve user abandonment issues [9]. All this information can provide a model of what preventative telehealth technologies should look like.

2.2 Telehealth for Older Adults

Telehealth is healthcare delivered remotely, often with the support of self-tracking technology and ubiquitous computing [22]. Telehealth is distinct from the Quantified Self for a few reasons: (1) data collected is specifically used for prevention or maintenance of

health conditions; (2) data collected is not typically widely shared, and is relayed only to health professionals, and (3) data collected is not personally analyzed; rather, it is reviewed by health professionals and care teams [20]. Additionally, this particular area of self-tracking has been progressively cultivated to cater to the demands of an ageing population. The majority of telehealth literature has largely focused on aiding management of chronic health conditions [1] [3] with a smaller subset which has focused on preventative telehealth. Although prior work on condition-managing telehealth can be insightful when designing preventative telehealth systems [14], users' motivations between categories largely differ. Preventative telehealth requires a significant amount of intrinsic desire to take care of one's health, whereas chronic telehealth often invokes a higher priority, and can be a part of a larger treatment plan. Furthermore, preventative telehealth is considerably more patient-driven than chronic telehealth tools. In this sense, preventative telehealth is more aligned with the self-tracking phenomena of the Quantified Self than it is with telehealth systems designed for chronic conditions. Mainly, in both preventative telehealth and Quantified Self, users adopt self-tracking technologies because they want to, not necessarily because they must.

Given all of this, we propose leveraging prior work that identifies the successes and pitfalls of the Quantified Self to help inform the design of preventative telehealth tools. We expect that this will help elicit synergistic interactions between patient and the doctor-mediated health monitoring apps.

3 DESIGN REQUIREMENTS

We initialized our design process by reflecting on prior work related to the Quantified Self and telehealth technologies for older adults. More specifically, we evaluated the strengths and weaknesses of the Quantified Self as discussed in the literature to see how we could adopt it for preventative telehealth technologies. In conjunction with this secondary research, we evaluated mobile applications designed for older adults to get a better sense of design elements and user experience. Furthermore, we reflected on the informational needs of older adults as users, and determined what data should be acquired for preventative measures. From our findings, we determined three design requirements that best fulfill user needs in preventative patient-driven telehealth: (1) increasing motivation and engagement (2) simplified health data input and management, and (3) trusted communication.

3.1 Increasing Motivation and Engagement

In order to be successful, preventative telehealth requires patients to be responsible for its adoption and continual use. The nature of preventative healthcare requires prolonged use without a specified timeline, which can result in tracking fatigue, one of the noted causes of abandonment in Quantified Self studies. Researchers have suggested numerous methods to mitigate or solve this issue, such as the use of rewards and creation of an unobtrusive and welcoming experience for the user [12] [26]. More specifically, Choe et. al assert that focusing on the positive makes the tracking experience more rewarding and less of a burden [8] [9]. Our designs implement these ideas to increase user motivation and engagement. Users are rewarded with positive language, gold stars and updates on their

“streak” upon successful completion of routine eye examinations. Interface design elements such as color, font and feel were taken into consideration to promote an inviting experience. Broadly, this requirement of increasing motivation and engagement was embodied by the simplicity of the application. By helping older adult users to easily learn the features and feel in control while navigating, we create a synergetic relationship between them and the app, which can increase chances of prolonged use.

3.2 Simplified Health Data Input and Management

Aggregating relevant healthcare information for older adults is helpful in goal setting, self-monitoring and memory retrieval. Research shows that users prefer simplicity in their self-managed data input [26], a finding that is particularly relevant for our target demographic. Quantified Self studies have suggested that simplifying or automating data upload, input and management, would greatly incentivize users to continue tracking their health [9]. As such, we streamlined eye capture to be taken directly on the smartphone application and uploading that data to a doctor to be as simple as possible. Furthermore, the development of a file feature provides patients with an overview of relevant details that support their preventative healthcare journey.

3.3 Trusted Data Analyzation and Communication

The increasing pervasiveness of technology has raised concerns for older adults about telehealth systems within their homes [24]. Studies by McNeil have identified distinct reasons for older adults' concern over privacy in health-monitoring systems which include (but are not limited to) self-protection, autonomy, and confidence [21]. Self-protection concerns the disclosure of sensitive information, autonomy is the ability to make independent decisions, and confidence is the control over what extent of information is disclosed, and to whom it is disclosed. However, specifically in terms of sharing data with health professionals, D'Haeseleer et. al showed that most participants were in favour of sharing their health-related data with a health-professional and believed that shared insights would only be beneficial since they were already in charge of their health status, [11]. Since evaluation of data by a healthcare professional is essential in telehealth, a system design that users trust is imperative. By creating a secure onboarding experience and allowing for communication with a predetermined optometrist, users can gain comfort in knowing who is viewing their information. Additionally, this externalized data evaluation eradicates the hindrances that arise when self-reflection on personal data is necessary, which was a previously identified flaw of the Quantified Self [12].

4 DESIGN OF SEEK

Prior work and design requirements were taken into careful consideration when designing the Seek mobile application. The interface design was intentionally created to support difficulties older adults notably face. For example, older adults can experience cognitive decline or memory loss and distorted color perception of cool tones. To address these needs, our user interface has captions at the top of every screen to provide user context of their navigation within

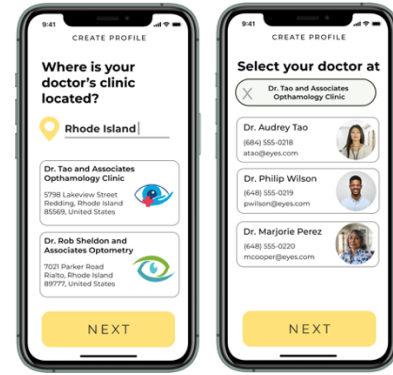


Figure 2: The two-step selection of the users' personal vision care specialist, as part of the onboarding process

the user experience, a warm tone colour palette, large call to action buttons that easily guide the user, and a single home button to replace a typical navigation panel in order to reduce likelihood of misinterpretations that derive from iconography and small fonts. We iteratively designed sketches, wireframes and two high-fidelity prototypes before settling on our presented mobile prototype. Our final design of Seek contains four main features to support preventative vision care: Onboarding, Instructions, Communication and Patient File. We discuss each of these next.

4.1 Onboarding

The onboarding process contributes to the user's feelings of safety and trust while interacting with the platform. When the user first encounters Seek for the first time, they will only be asked to fill in basic information such as their name and phone number. Subsequently, users search for their optometrist directly on the mobile app, in order to initiate a mutual connection. The search process for their doctor is a simple two-step process. First, users narrow down their search by finding the clinic their doctor is associated with, either by name or location, and secondly, they will select their registered optometrist at their specified clinic (Figure 2).

To promote self-protection and trust, our design for onboarding requires disclosure of sensitive information only after both the user and their optometrist have verified and accepted the connection. Additionally, the mutual verification of the connection was designed so that the doctors themselves can partially assist their patients in transitioning to a virtual assessment. This meets our design requirements, as it further reduces privacy concerns and increases the patients' trust and confidence that their data is being analyzed by their doctor only.

4.2 Instructions for Image Capture

Older adults find it difficult to learn and adopt new technologies, especially when they have not been previously exposed to similar patterns in design or functionality [7] [25]. This is certainly important in our system, considering the novelty of the device's design and functionality. To aid successful adoption of the paired ocular device that is essential to successful ocular image capture, we've included thorough step-by-step instructions with visuals on how

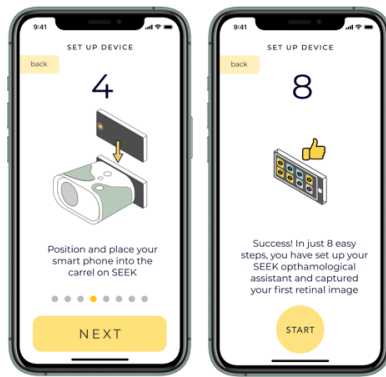


Figure 3: A step-by-step instructional guide to help users adopt the entire system.

to use the device and upload an ocular image to their physician (Figure 3). This feature is integral to the entire system because it provides users the skills required to succeed in adopting the technology. Users who are confident in their capabilities can benefit from the use of the system and earn rewards and gold stars, which will further incentivize them to maintain use.

4.3 Communication with Doctor

As previously mentioned, simplified data input has been suggested as a way to promote long term use of self-tracking technologies [8] [9] [12]. Inspired by this, the Seek application is designed so that users are only allowed to upload images to their own doctor or request a call, no text-based correspondence necessary (Figure 4). Patients simply click ‘upload photo’ where they can then preview previous captures or take new photos, click an image to preview, and finally send them directly to their doctor. This unimodal form of communication where users only upload ocular images and nothing else, allows the user the liberty to not be overwhelmed by, or overthink their input of data – which in this case is correspondence with their doctor. Additionally, this feature also aligns with the trusted communication design requirement, since users are protected from any possibility of accidental disclosure of sensitive information. If necessary, users can request a call from their doctors. We decided to include this feature because spoken communication can often reduce miscommunication, and the relative formality of a phone call can act as a screening threshold of importance for the user before a request is made.

4.4 My File

Lastly, we included a file feature where users can review relevant details that can aid their preventative vision care journey (Figure 5). Information such as their condition, doctor, recent check-ups and in-person appointment are found here. This feature meets our first two design requirements which as it aims to increase user motivation with words of affirmation and positivity as well as providing information management to facilitate the user experience. While the entire design of the system was meant to support older adults, this feature is particularly helpful. Older adults struggle may with memory, active recall and other aspects related to cognitive decline

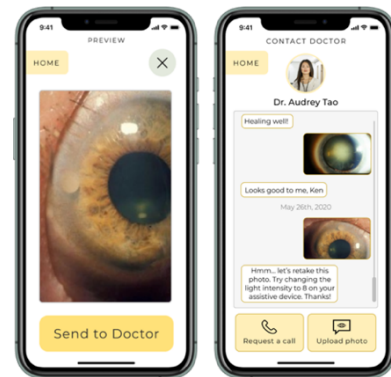


Figure 4: Patients exclusively communicate with their doctor through ocular images.



Figure 5: The file feature includes relevant details and visualization of the user’s progress.

[7]. This simple context reminders of their goals and progress can be useful and assuring, but also crucial for periodic preventative care maintenance.

5 USER SCENARIOS

Two user scenarios were created to explore realistic experiences older adults might face with the adoption of Seek.

Scenario 1: Lily is a vibrant 78-year-old lady who suffers from Alzheimer’s disease, and has shown early signs of Glaucoma. She receives a notification from her Seek app, suggesting she upload her bi-weekly retinal image for her optometrist. After uploading and sending her retinal image to her doctor, she briefly views her file. Seeing a reminder that she’s due for an annual in-person check-up, she requests a call from her optometrist directly through the app and continues going about her day.

Scenario 2: Henry lives in a rural countryside with his wife who is diagnosed with terminal cancer. While he’s been busy taking care of his wife’s health, his own health has been deprioritized, and non-urgent visits to the city are at the bottom of his to-do list. One of the tools he uses to manage his health is Seek. Every other Monday, he powers up his device and captures an image to send to his optometrist. It takes less than 10 minutes to capture and upload,

and then he waits for a notification within the week that his doctor has reviewed his image. He enjoys the positive affirmations he receives when he's on a streak. It automatically lifts his mood, and the positive embodied feeling always motivates him to check-in again.

6 DISCUSSION AND CONCLUSION

The system design presented in this paper describes a potential solution to support older adults in their pursuit of preventative vision care. Seek offers the community an alternative way to get the professional opinion of their vision care specialist, without having to worry about certain aspects of accessibility involved with traditional optometry. Due to the need for inherent user motivation required for successful adoption and prolonged use, we adapted findings from the Quantified Self. Research about this self-tracking movement thoroughly describes user motivations for using self-tracking technologies and quantifying personal data, as well as strategies to overcome the studied pitfalls of long-term adoption that lead to abandonment.

This paper has focused on describing the design requirements, ideation and prototype of the mobile application of our system. While this system is not meant to replace in-clinic vision examinations entirely, it does aim to inspire conversation, imagination and research in the broader HCI community – relating to the advancement of current technologies and design frameworks that have the power to enhance telehealth tools. To support this, our next steps include an evaluation of our mobile application with older adults to further explore usage and behavioural patterns with preventative telehealth technology.

We believe that adoption of preventative telehealth will be imperative for the future, as the aging population quickly increases in size. While our system has specifically applied our design requirements to offer a solution for age-related vision loss prevention, such requirements can be equally useful in support of other preventative health measures. These applications may also be used beyond the currently intended audience of older adults, such as those living in rural populations who share similar barriers to accessible healthcare.

ACKNOWLEDGMENTS

Thank you to Professor Karyn Moffatt and Dr. Carolyn Pang in their continual guidance and support for this research, to Xue Ting Song for inspiring me to undertake this project, and the CHI 2021 reviewers for their time. Lastly, a big thank you to the broader HCI community for contributing inspirational literature to this field that has the potential to change many lives for the better.

REFERENCES

- [1] Adelson, J., Bourne, R. R. A., Briant, P. S., Flaxman, S., Taylor, H., Jonas, J. B., et al., Causes of blindness and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to VISION 2020: the Right to Sight: an analysis for the Global Burden of Disease Study. *Lancet Global Health*. 2020.
- [2] Norman Archer and Reza Aria, "Reducing Risk from Chronic Illness with Life Style Changes Supported by Online Health Self-Management," *2019 IEEE/ACM 1st International Workshop on Software Engineering for Healthcare (SEH)*, Montreal, QC, Canada, 2019, pp. 73–76, doi: 10.1109/SEH.2019.00020.
- [3] Tobenna D. Anekwe and Ilya Rahkovsky, 2018. Self-Management: A Comprehensive Approach to Management of Chronic Conditions. *American Journal of Public Health* **108**, S430_S436, <https://doi.org/10.2105/AJPH.2014.302041r>
- [4] Amritpal Singh Bhachu, Nicolas Hine, and John Arnott. 2008. Technology devices for older adults to aid self management of chronic health conditions. In *Proceedings of the 10th international ACM SIGACCESS conference on Computers and accessibility (Assets '08)*. Association for Computing Machinery, New York, NY, USA, 59–66. DOI:<https://doi-org.proxy3.library.mcgill.ca/10.1145/1414471.1414484>
- [5] Frank Bentley, Konrad Tollmar, Peter Stephenson, Laura Levy, Brian Jones, Scott Robertson, Ed Price, Richard Catrambone, and Jeff Wilson. 2013. Health Mashups: Presenting Statistical Patterns between Wellbeing Data and Context in Natural Language to Promote Behavior Change. *ACM Trans. Comput.-Hum. Interact.* **20**, 5, Article 30 (November 2013), 27 pages. DOI:<https://doi.org/10.1145/2503823>
- [6] Shlomo Berkovsky, Mac Coombe, Jill Freyne, Dipak Bhandari, and Nilufar Baghaei. 2010. Physical activity motivating games: virtual rewards for real activity. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. Association for Computing Machinery, New York, NY, USA, 243–252. DOI:<https://doi-org.proxy3.library.mcgill.ca/10.1145/1753326.1753362>
- [7] Alethea Blackler, Doug Mahar, and Vesna Popovic. 2010. Older adults, interface experience and cognitive decline. In *Proceedings of the 22nd Conference of the Computer-Human Interaction Special Interest Group of Australia on Computer-Human Interaction (OZCHI '10)*. Association for Computing Machinery, New York, NY, USA, 172–175. DOI:<https://doi-org.proxy3.library.mcgill.ca/10.1145/1952222.1952257>
- [8] Eun Kyoung Choe, Bongshin Lee, Matthew Kay, Wanda Pratt, and Julie A. Kientz. 2015. SleepTight: low-burden, self-monitoring technology for capturing and reflecting on sleep behaviors. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. Association for Computing Machinery, New York, NY, USA, 121–132. DOI:<https://doi.org/10.1145/2750858.2804266>
- [9] Eun Kyoung Choe, Nicole B. Lee, Bongshin Lee, Wanda Pratt, and Julie A. Kientz. 2014. Understanding quantified-selfers' practices in collecting and exploring personal data. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1143–1152. DOI:<https://doi.org/10.1145/2556288.2557372>
- [10] Pooja M. Desai, Elliot G. Mitchell, Maria L. Hwang, Matthew E. Levine, David J. Albers, and Lena Mamykina. 2019. Personal Health Oracle: Explorations of Personalized Predictions in Diabetes Self-Management. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, Paper 370, 1–13. DOI:<https://doi.org/10.1145/3290605.3300600>
- [11] Ine D'Haeseleer, Kathrin Gerling, Dominique Schreurs, Bart Vanrumste, and Vero Vanden Abeele. 2019. Ageing is Not a Disease: Pitfalls for the Acceptance of Self-Management Health Systems Supporting Healthy Ageing. In *The 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19)*. Association for Computing Machinery, New York, NY, USA, 286–298. DOI:<https://doi.org/10.1145/3308561.3353794>
- [12] Daniel A. Epstein, An Ping, James Fogarty, and Sean A. Munson. 2015. A lived informatics model of personal informatics. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. Association for Computing Machinery, New York, NY, USA, 731–742. DOI:<https://doi.org/10.1145/2750858.2804250>
- [13] Henner Gimpel, Marcia Nißen, Roland Görlitz. (2013). Quantifying the Quantified Self: A Study on the Motivations of Patients to Track Their Own Health. *International Conference on Information Systems (ICIS 2013): Reshaping Society Through Information Systems Design*. 4.
- [14] Casandra Grundstrom, Violeta Gaveikaite, Louise Barkhuus, Nicos Maglaveras, Stefan Winter, and Joanna Chouvarda. 2019. User Profiles of Healthcare Professionals in a Chronic Obstructive Pulmonary Disease Care Pathway. In *Proceedings of the 13th EAI International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth'19)*. Association for Computing Machinery, New York, NY, USA, 444–451. DOI:<https://doi.org/10.1145/3329189.3329243>
- [15] Gimpel, Henner & Nißen, Marcia & Görlitz, Roland. (2013). Quantifying the Quantified Self: A Study on the Motivations of Patients to Track Their Own Health. *International Conference on Information Systems (ICIS 2013): Reshaping Society Through Information Systems Design*. 4.
- [16] Mick Jongeling, Dan Xu, Ilaria Zonda, and Gijis Huisman. 2019. Opportunities of Quantified Self for Resocialisation of (Ex-)Convicts. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems (CHI EA '19)*. Association for Computing Machinery, New York, NY, USA, Paper LBW1318, 1–6. DOI:<https://doi.org/10.1145/3290607.3312885>
- [17] Rohit Ashok Khot, Ryan Pennings, and Florian 'Floyd' Mueller. 2015. EdiPulse: Supporting Physical Activity with Chocolate Printed Messages. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '15)*. Association for Computing Machinery, New York, NY, USA, 1391–1396. DOI:<https://doi-org.proxy3.library.mcgill.ca/10.1145/2702613.2732761>
- [18] Amanda Lazar, Christian Koehler, Theresa Jean Tanenbaum, and David H. Nguyen. 2015. Why we use and abandon smart devices. In *Proceedings of the*

- 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15). Association for Computing Machinery, New York, NY, USA, 635–646. DOI:<https://doi.org/10.1145/2750858.2804288>
- [19] Ian Li, Anind K. Dey, and Jodi Forlizzi. 2011. Understanding my data, myself: supporting self-reflection with ubicomp technologies. In Proceedings of the 13th international conference on Ubiquitous computing (UbiComp '11). Association for Computing Machinery, New York, NY, USA, 405–414.
- [20] Ian Li, Anind Dey, and Jodi Forlizzi. 2010. A stage-based model of personal informatics systems. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10). Association for Computing Machinery, New York, NY, USA, 557–566. DOI:<https://doi.org/10.1145/1753326.1753409>
- [21] Gastón Márquez, Hernán Astudillo, and Carla Taramasco. 2019. Exploring security issues in telehealth systems. In Proceedings of the 1st International Workshop on Software Engineering for Healthcare (SEH '19). IEEE Press, 65–72. DOI:<https://doi-org.proxy3.library.mcgill.ca/10.1109/SEH.2019.00019>
- [22] Merriel, S. W., Andrews, V., & Salisbury, C. (2014). Telehealth interventions for primary prevention of cardiovascular disease: a systematic review and meta-analysis. *Preventive medicine*, 64, 88–95. <https://doi.org/10.1016/j.ypmed.2014.04.001>
- [23] Andrew McNeill, Pam Briggs, Jake Pywell, and Lynne Coventry. 2017. Functional privacy concerns of older adults about pervasive health-monitoring systems. In Proceedings of the 10th International Conference on Pervasive Technologies Related to Assistive Environments (PETRA '17). Association for Computing Machinery, New York, NY, USA, 96–102. DOI:<https://doi.org/10.1145/3056540.3056559>
- [24] Francisco Nunes, Nervo Verdezoto, Geraldine Fitzpatrick, Morten Kyng, Erik Grönvall, and Cristiano Storni. 2015. Self-Care Technologies in HCI: Trends, Tensions, and Opportunities. *ACM Trans. Comput.-Hum. Interact.* 22, 6, Article 33 (December 2015), 45 pages. DOI:<https://doi.org/10.1145/2803173>
- [25] Marita A. O'Brien, Wendy A. Rogers & Arthur D. Fisk. 2012. Understanding age and technology experience differences in use of prior knowledge for everyday technology interactions. *ACM Transactions on Accessible Computing*, 4(2), 1–27. doi:10.1145/2141943.2141947
- [26] Jeungmin Oh and Uichin Lee, "Exploring UX issues in Quantified Self technologies," *2015 Eighth International Conference on Mobile Computing and Ubiquitous Networking (ICMU)*, Hakodate, 2015, pp. 53–59, doi: 10.1109/ICMU.2015.7061028.
- [27] Amon Rapp, Federica Cena, Judy Kay, Bob Kummerfeld, Frank Hopfgartner, Till Plumbaum, Jakob Eg Larsen, Daniel A. Epstein, and Rüben Gouveia. 2016. New frontiers of quantified self 2: going beyond numbers. In Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct (UbiComp '16). Association for Computing Machinery, New York, NY, USA, 506–509. DOI:<https://doi-org.proxy3.library.mcgill.ca/10.1145/2968219.2968331>
- [28] John Rooksby, Mattias Rost, Alistair Morrison, and Matthew Chalmers. 2014. Personal tracking as lived informatics. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14). Association for Computing Machinery, New York, NY, USA, 1163–1172. DOI:<https://doi.org/10.1145/2556288.2557039>
- [29] United Nations 2019. World Population Ageing 2019 Highlights. Retrieved January 11, 2021 from <https://www.un.org/en/development/desa/population/publications/pdf/ageing/WorldPopulationAgeing2019-Highlights.pdf>
- [30] Alishia D. Williams. 2016. Harnessing the quantified self movement for optimal mental health and wellbeing. In Proceedings of the first Workshop on Lifelogging Tools and Applications (LTA '16). Association for Computing Machinery, New York, NY, USA, 37. DOI:<https://doi.org/10.1145/2983576.2983585>
- [31] WHO 2019. World report on vision. Retrieved January 11, 2021 from <https://www.who.int/publications-detail-redirect/world-report-on-vision>